

REPORT

Real-World Retail Sales Analysis Project

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Objective

The objective of this project is to analyze retail sales data and identify trends, profit patterns, and business insights using data analytics and visualization techniques.

Dataset

Sample Superstore Retail Dataset

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook
- Scikit-Learn

Dataset Information

```
12]: print(df.info())

<class 'pandas.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Row ID                 9994 non-null  int64  
1   Order ID               9994 non-null  str     
2   Order Date             9994 non-null  str     
3   Ship Date              9994 non-null  str     
4   Ship Mode              9994 non-null  str     
5   Customer ID            9994 non-null  str     
6   Customer Name          9994 non-null  str     
7   Segment                9994 non-null  str     
8   Country                9994 non-null  str     
9   City                   9994 non-null  str     
10  State                  9994 non-null  str     
11  Postal Code            9994 non-null  int64  
12  Region                 9994 non-null  str     
13  Product ID             9994 non-null  str     
14  Category               9994 non-null  str     
15  Sub-Category           9994 non-null  str     
16  Product Name           9994 non-null  str     
17  Sales                   9994 non-null  float64 
18  Quantity               9994 non-null  int64  
19  Discount                9994 non-null  float64 
20  Profit                  9994 non-null  float64 
dtypes: float64(3), int64(3), str(15)
memory usage: 1.6 MB
None
```

Figure 1: Dataset Information

Explanation:

This figure provides details about the dataset including the number of records, columns, data types, and available business attributes such as sales, profit, category, and region.

Statistical Summary

```
[13]: print(df.describe())
```

	Row ID	Postal Code	Sales	Quantity	Discount \
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	4997.500000	55190.379428	229.858001	3.789574	0.156203
std	2885.163629	32063.693350	623.245101	2.225110	0.206452
min	1.000000	1040.000000	0.444000	1.000000	0.000000
25%	2499.250000	23223.000000	17.280000	2.000000	0.000000
50%	4997.500000	56430.500000	54.490000	3.000000	0.200000
75%	7495.750000	90008.000000	209.940000	5.000000	0.200000
max	9994.000000	99301.000000	22638.480000	14.000000	0.800000

	Profit
count	9994.000000
mean	28.656896
std	234.260108
min	-6599.978000
25%	1.728750
50%	8.666500
75%	29.364000
max	8399.976000

Figure 2: Statistical Summary

Explanation:

The statistical summary presents key measures such as mean, minimum, maximum, and standard deviation for sales, profit, quantity, and discount.

Sales by Category

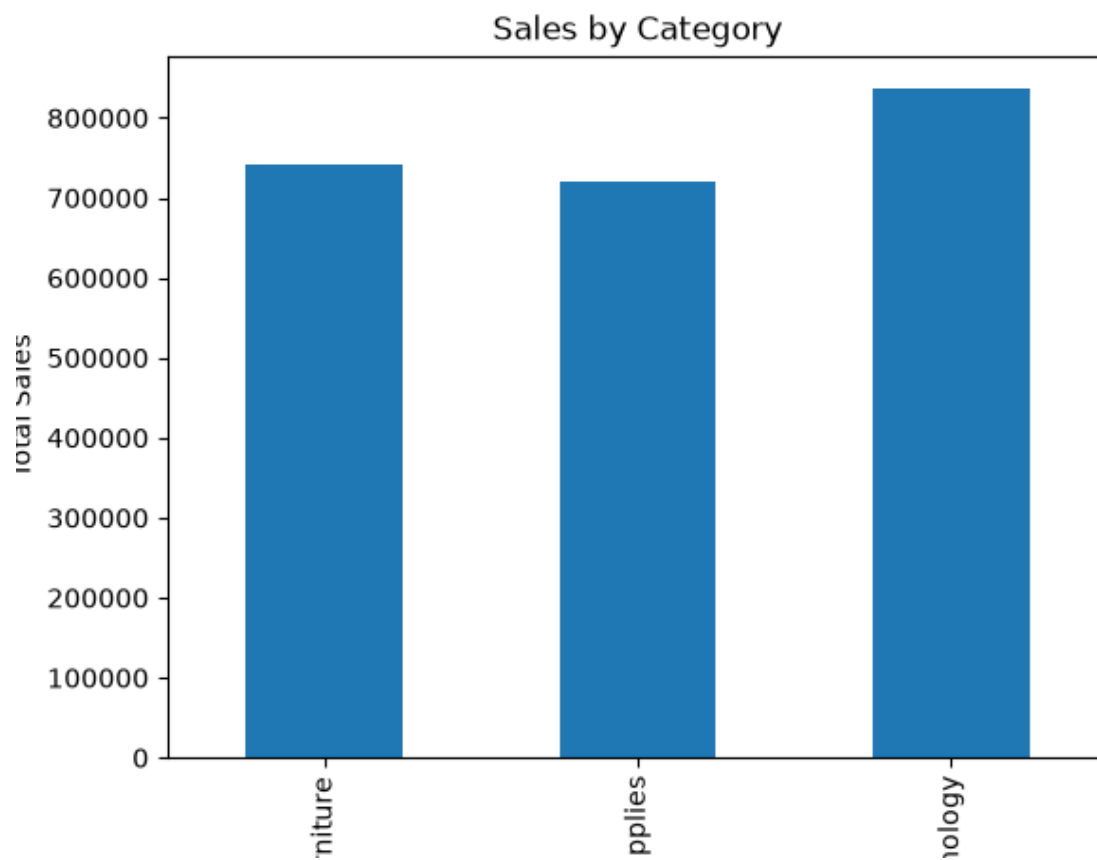


Figure 3: Sales by Category

Explanation

This chart compares total sales across different product categories.

Observation

- Some categories contribute significantly more revenue than others.
- Category performance helps identify major revenue sources.

Profit by Category

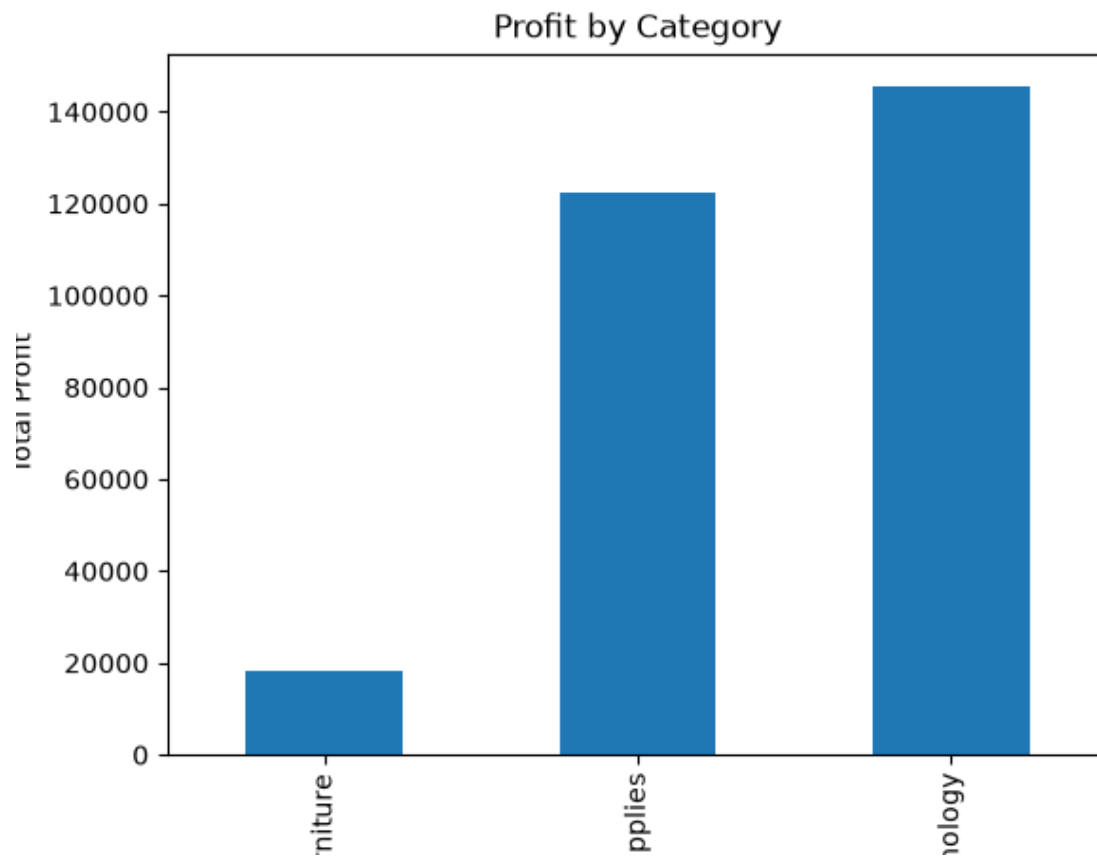


Figure 4: profit by category

Explanation

This chart illustrates profit generated by each product category.

Observation

- High sales do not always result in high profit.
- Profitability differs across categories.

Sales by Region

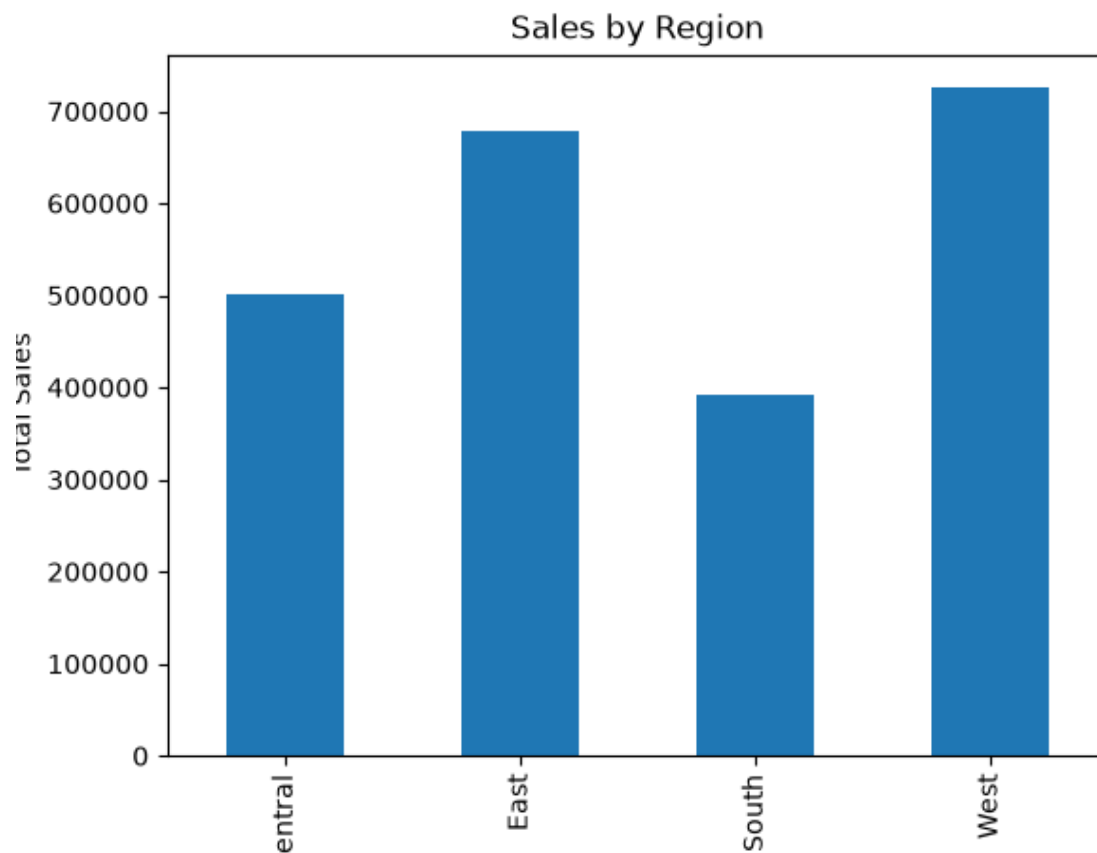


Figure 5: sales by region

Explanation

This chart compares total sales across different regions.

Observation

- Certain regions generate higher sales than others.
- Regional analysis helps identify strong and weak markets.

Top 10 Products by Sales

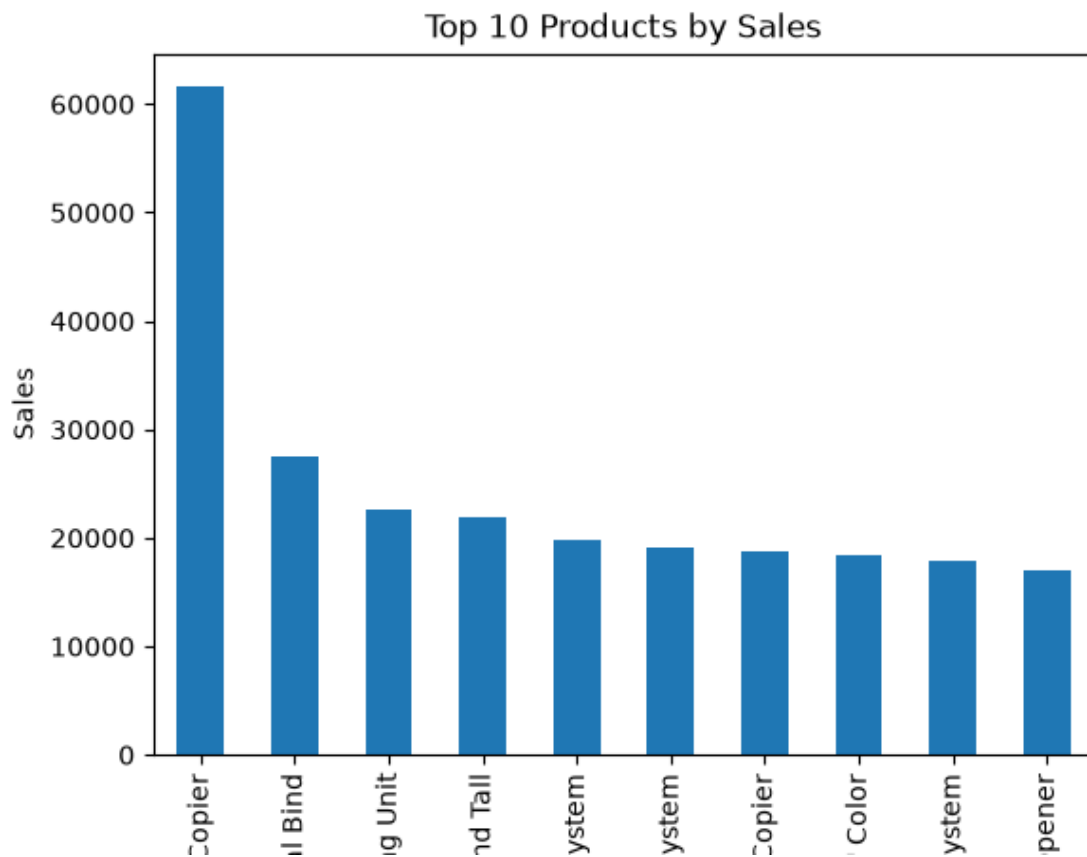


Figure 6: top10 products

Explanation

This visualization highlights the products contributing the highest sales revenue.

Observation

- A small number of products contribute significantly to overall sales.
- Top-performing products can be prioritized for business growth

Conclusion

Key Findings

1. Sales performance varies across categories.
2. Profitability differs among product categories.
3. Regional sales analysis reveals market trends.
4. Top products contribute a major share of revenue.
5. Data analytics helps support business decision-making.

Conclusion

This project successfully analyzed retail sales data using Python and data visualization techniques. The analysis provided valuable insights into category performance, profitability, regional sales trends, and top-selling products. These findings can help businesses make informed decisions and improve overall performance.